Congestion-aware clustering

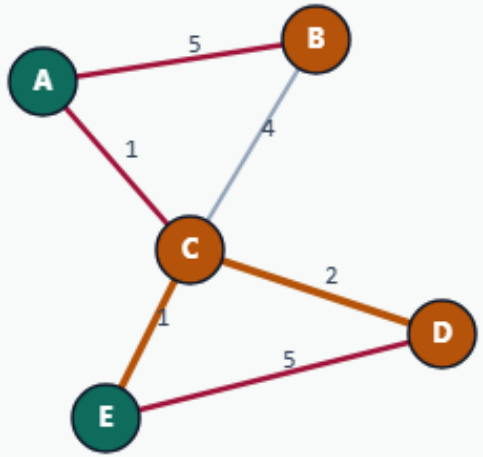
When bridge edges are congested, cutting them becomes expensive

Start from BAD_SEED

CONGESTION-AWARE CLUSTERING

Start from BAD_SEED

Step 1 / 5 · bad-seed · module04-01-congestion-aware-clustering



Explanation

Same cutsize-12 seed AE|BCD. Congestion map marks C–D=5 and C–E=4 — cheap wire cuts may be expensive for routing.

- plain cut ignores congestion
- penalty sums cong on cut edges
- λ trades plain vs penalty

```
seed cut: 12
cong: C|D=5, C|E=4
```

Start from BAD_SEED

λ=0 → ordinary FM

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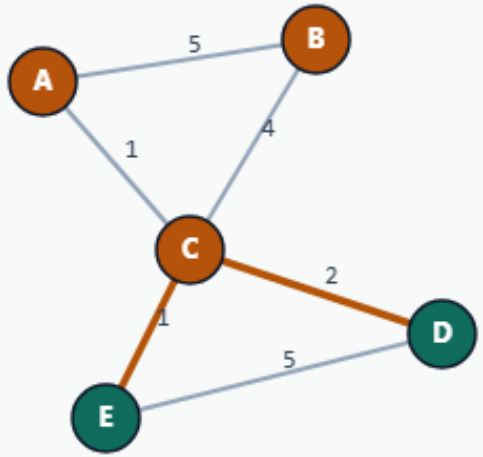
λ=0 → ordinary FM

Step 2 / 5 · lam0 · module04-01-congestion-aware-clustering



Explanation

With λ=0, boosted weights equal original weights. FM recovers ABC|DE: plain=3 but penalty=9 because both congested bridges are cut.

- plain=3, pen=9
- parts: ABC|DE
- Objective ignores routing pain

```
λ=0
plain=3
pen=9
```

λ=0 → ordinary FM

Boost weights by λ·cong

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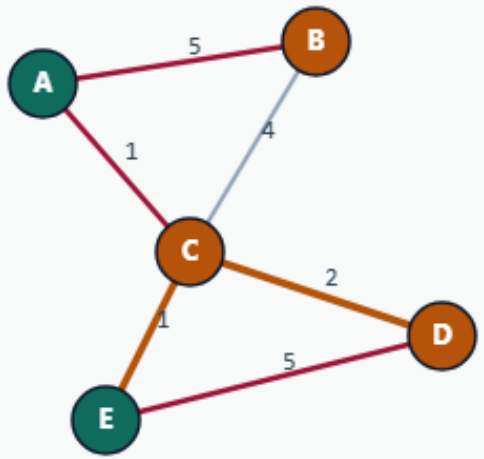
Boost weights by λ·cong

Step 3 / 5 · boost · module04-01-congestion-aware-clustering



Explanation

For λ=5, C–D and C–E become very expensive to cut. FM optimizes the boosted graph, then we report plain cut and congestion penalty separately.

- w' = w + λ·cong
- Run FM on w'
- Score plain + λ·pen on original

```
λ=5
C–D and C–E heavily boosted
```

Boost weights by λ·cong

λ=5 → pen 0, plain 5

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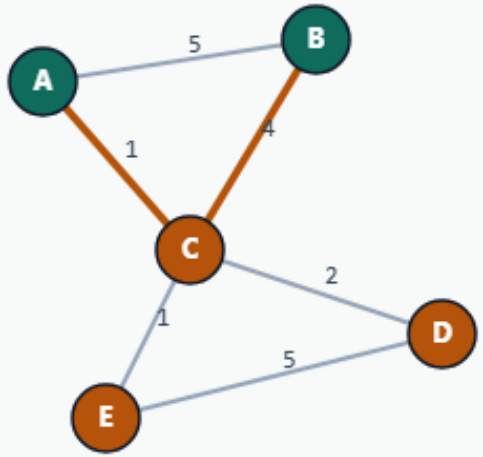
λ=5 → pen 0, plain 5

Step 4 / 5 · lam5 · module04-01-congestion-aware-clustering



Explanation

Result AB|CDE: plain cut rises to 5, but congestion penalty drops to 0 — congested bridges stay internal.

- parts: AB|CDE
- plain=5, pen=0
- objective=5

```
λ=5
plain=5
pen=0
```

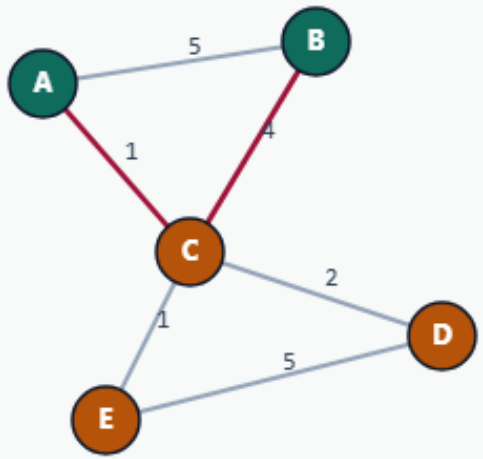
λ=5 → pen 0, plain 5

Objective tradeoffs

CONGESTION-AWARE CLUSTERING

Objective tradeoffs

Step 5 / 5 · takeaway · module04-01-congestion-aware-clustering



Explanation

EDA flows rarely optimize cut alone. λ makes the congestion tax explicit so students see the Pareto move between wire and routing.

- λ=0 favors classic communities
- λ>0 protects congested edges
- Same FM engine, different weights

```
Goldens: (3,9) vs (5,0)
```

Objective tradeoffs

Browser lab track

- In the browser lab, compare lambda zero with lambda five from the same bad seed
- Watch the penalty drop to zero when the congested bridge is avoided

Implement track

- Run congestion-aware partition at lambda zero and five
- Confirm plain cut three with penalty nine at lambda zero

Implement track — try these

```
export PYTHONPATH=../common

python ../common/solvers.py examples/tiny_graph.json --mode congestion
--seed ../module02-05-kernighan-lin/examples/seed_partition.json
--congestion examples/congestion.json --lambda 5
```

Pitfalls to watch

- Applying congestion only after FM misses the point, weights must change before moves
- Huge lambda can ignore connectivity entirely

Your turn

- Match both lambda goldens